

Design and Implementation of Real-Time Pump Failure Analysis and Monitoring System By Using ESP-32 and IoT Sensors

Sanket Gajanan Barhate
Dept of Mechanical
Engineering,
Sinhgad College Of
Engineering, Vadgaon,
Pune, India
sanketbarhate3@gmail.com

Aayush Anilkumar Sharma
Dept of Mechanical
Engineering,
Sinhgad College Of
Engineering, Vadgaon,
Pune, India
sharmaayush0803@gmail.com

Shubham Pravin Chopade
Dept of Mechanical
Engineering,
Sinhgad College Of
Engineering, Vadgaon,
Pune, India
shubhamchopade440@gmail.com

Chaitanya Rajendra Akolkar
Dept of Mechanical
Engineering,
Sinhgad College Of
Engineering, Vadgaon,
Pune, India
chaitanyaakolkar75@gmail.com

Ram Shivaji Mote
Dept of Mechanical Engineering,
Sinhgad College Of Engineering, Vadgaon,
Pune, India
rsote.scoe@sinhgad.edu

Sanchit Babarao Dhotre
Dept of Mechanical Engineering,
Sinhgad College Of Engineering, Vadgaon,
Pune, India
dhotesanchit@gmail.com

Abstract—This paper will outline an IoT-based ESP32-based real-time pump failure detection and monitoring system designed and implemented using sensors. The industrial and domestic pumps cannot be overlooked because when the pump fails, it may result in high operational costs, maintenance costs and the lack of availability of the pump. An intelligent monitoring system that includes temperature, and flow sensors to check the working conditions of the pump is the proposed solution to the given dilemma. The system proposed is founded on the ESP32 microcontroller, which offers real-time data capture, processing, and wireless communication. The sensor data are analyzed to predict anomalies, including overheating changes and abnormal flow patterns, that could be signs of pump failures. The system will be automated in such a way that the pump will be turned off when the conditions are severe due to a switch button (relay) that would not allow additional damage. It will be cost-effective, small, and appropriate for small and medium-sized applications. The experimental outcomes show that the system may be utilized to control the essential parameters in real-time and to identify defects at the initial stages, which can enhance the reliability of pumps and their performance. The IoT remote monitoring can also be applied in the process of the implementation, and it provides the possibility of real-time supervision and predictive maintenance. The provided work will assist in designing intelligent pump monitoring tools that will make use of the collaboration of embedded systems and IoT technologies, which will provide a viable alternative to reduce downtime and extend the working schedule of the equipment.

Keywords: *ESP32, Pump Failure Detection, Internet of Things (IoT), Real-Time Monitoring, Predictive Maintenance, Sensor-Based System, Data Acquisition, Centrifugal Pump*

I. INTRODUCTION

Industrial, agricultural and domestic use are examples of typical centrifugal pumps, necessitating the movement of fluids, such as supply, irrigation, waste treatment, and oil and gas utilities. Pumps constantly run, and the working conditions are not the same; therefore, they can easily fail (cavitation, bearing damage, impeller wear, overheating and shaft misalignment). Such failures may cause unwanted downtime, higher maintenance costs, and loss of efficient operation [1], [2].

The last few years have seen a lot of research on pump health prognostics and fault anticipation. The concept behind predictive maintenance is to embrace more data-driven sensor-based monitoring systems to enhance its efficiency. It has also been demonstrated that with the help of smart sensors based on Internet of Things (IoT) technologies, it becomes possible to measure the parameters of pumps (vibration, temperature, flow rate) almost in real time, which enhances the performance and reliability of the machine [3], [4]. Besides, machine learning and artificial intelligence algorithms have been applied to sensor data to obtain accurate fault prediction and classification to make condition monitoring systems more useful [5].

Various methods of diagnosing pump failures have been discussed by a number of authors. It has been suggested that hybrid systems with a mixture of physics and data-driven systems be employed to enhance accuracy and interpretability in fault detection [6]. It is also found that monitoring systems based on IoT and employing both wireless and wired sensors may provide highly accurate real-time diagnostics [7]. In addition, vibration analysis techniques, such as wavelet transforms and neural networks, have been effectively applied in fault detection in their early stages, such as cavitation and bearing faults [8], [9]. The strategies observe the importance of effective data acquisition and processing for predictive maintenance.

Regardless of these developments, certain systems available exist that are either complex, costly or hard to use in small and medium-scale applications. It should provide high-quality and real-time monitoring and early fault detection at low cost, a small and easy-to-install solution. The appearance of embedded systems also implies the use of microcontrollers like ESP32, which has an in-built Wi-Fi to make it easy to acquire and process data, and conduct remote monitoring. The given work is dedicated to the design and implementation of a real-time pump failure detector and monitoring system based on the ESP32 microcontroller and a set of sensors, such as a temperature sensor, and a flow sensor. The system is able to constantly gather operational values, identify abnormal conditions, and turn into a relay to switch off the pump during an emergency. The purpose of the suggested system is to ensure that the pump is more reliable, less expensive to maintain, and more efficient in its work with the help of IoT. Intelligent and scalable pump-monitoring solutions are emerging as a consequence of the current work and can be applied in industrial practice.

The proposed pump failure monitoring system is a set of continuous measurements of different parameters of operation that are measured on a regular basis using various sensors to check the status of a pump. The parameters are chosen due to their relevance to the identification of the common malfunctions of the hand pumps in the case of overheating, cavitation imbalance. There will be three important independent variables in the system, which include temperature, and flow rate and are to be measured with the help of the relevant sensors and calculated with the help of the ESP32 microcontroller. These parameters can be used to identify abnormal operating conditions; thus, fault detection can be performed early.

II. LITERATURE REVIEW

Centrifugal pumps can be used in many industrial, agricultural and domestic applications, such as the supply system, irrigation, wastewater treatment, and the oil and gas industry. The pumps continuously tend to remain in motion and are subjected to varying conditions, which makes them very vulnerable to cavitation, damage, impeller wear, overheating, and misalignment. Such failures may cause unforeseen downtime, high maintenance costs, and poor performance [1], [2].

The past few years have seen a lot of research on the topic of pump health monitoring and fault detection. Predictive maintenance has been embraced using sensor-based monitoring systems to enhance effectiveness. It is also established that through the fusion of smart sensors and Internet of Things (IoT), it is possible to control the parameters of the pump (vibration, temperature, flow rate) almost in real time and, therefore, enhance the operation of the machine and make it more reliable [3], [4]. In addition, machine learning and artificial intelligence techniques of sensor data processing have been introduced to generate the correct fault prediction and classification, thus making condition monitoring systems more efficient [5].

Different scholars have suggested the use of varied techniques when diagnosing the faults of pumps. To enhance the accuracy and interpretability of fault detection, hybrid models, which relate physics-based and data-driven modelling, should be applied [6]. Real-time diagnostics using wireless and wired IoT-based sensor monitoring

systems have also been demonstrated to be very precise [7]. Further vibration analysis techniques, including wavelet transforms and neural networks, have been effectively applied to the early prediction of damage, e.g. cavitation and bearing damage [8], [9]. The strategies put emphasis on the significance of viable data acquisition and processing to predictive maintenance.

In spite of them, there are certain systems that are currently available and are either complicated, costly or hard to use in small and medium-scale applications. It should have a small, affordable, and effortless-to-set-up solution, which will provide real-time monitoring and fault early detection of high quality. Remote data acquisition, processing, and monitoring are made possible through the creation of embedded systems and microcontrollers, including the ESP32, which has built-in Wi-Fi connectivity. The project concentrates on the construction of a real-time pump failure detection and monitoring system on the basis of the ESP32 microcontroller and temperature, and flow sensors. Operation data is collected continuously, anomalies are detected, and a relay is automatically released that cuts off the pump when there is an emergency. The proposed system will strive to improve the reliability of the pump, lower the maintenance expenses, and increase the efficiency of the operations by diagnosing and monitoring the faults and utilizing the IoT. The existing study leads to smart pump-monitoring solutions, which can be extended to meet industry practice.

The suggested system of pump failure monitoring continuously controls the work of a pump by means of different sensors to determine the health of a pump. These parameters have been chosen depending on their usefulness in determining common errors of the hand-pump, which include overheating, and cavitation. Three key independent variables that will be taken into consideration in the system and measured with the help of the necessary sensors are the temperature, and flow rate, which the ESP32 microcontroller will manipulate. These parameters can be used to identify abnormal operating conditions and reveal the faults at an early stage.

The architecture of the smart home automation constructed by Ahmed et al. [11] is based on the ESP32 and sensors of the environment and automatic control. The proposed methodology will involve the installation of DHT22, MQ2, MQ135, soil moisture, and LDR sensors and a relay that will be used to automate them. The Wi-Fi-linked mobile app allows managing and controlling appliances, lighting, and irrigation systems in real time. The other safety systems provided are gas detectors and overheating. The findings reveal that intelligent homes have improved efficiency in terms of energy consumption, comfort and monitoring the environment. The system will be able to automate all functions in the house and alert about safety concerns in real time. Nevertheless, the fact that big buildings cannot be scaled, the poor use of AI in smart control, and the absence of research on cybersecurity-related problems are a research gap. It is also not discussed well that performance evaluation is a possibility in various real-life scenarios.

Imam and Ghosh [12] presented an ESP32-based real-time industrial-parameter monitoring system that is based on safety. Other sensors (gaseous and flame sensors, DHT11, and LDRs) are also incorporated into the style to control

environmental and dangerous situations. The ESP32 then reads sensor data and sends it wirelessly through Wi-Fi to a web-based dashboard to be visualized in real time. The notification system notifies about overweight parameters beyond the set limits, thus taking care of hazards early enough. The findings indicate that it gives viable real-time surveillance, excellent industrial safety and an effective gas-leakage and fire-hazard system as presented in the system architecture on page 1. The gap in the research, however, is that it does not have any predictive analytics that can identify faults, cannot be handled at scale within an industrial setting, and cannot be combined with machine learning to identify anomalies.

The smart healthcare monitoring system introduced by Abhang et al. [13] is based on the IoT-IV saline monitoring. It uses ESP32 microcontrollers, load cell sensors and GSMs to monitor the saline level and abnormalities in real-time. The data is transmitted to an online system, and the notification is sent through SMS and mobile applications in case of breach of critical levels. This system automates the monitoring of saline and minimizes human handoffs, enhancing patient safety. Findings have revealed increased efficiency in the medical operations, the minimization of human error and a faster response rate in the handling of emergency cases. The system perfectly fits to avoid the risks of saline depletion and backflow. Nevertheless, the study also has its limitations: it is not incorporated into the hospital management systems, can only be used in the multi-patient setting, and does not have developed predictive analytics to monitor the condition of patients.

In a study conducted by Suryowinoto et al. [14], the use of an intelligent predictive maintenance system is suggested using the IoT sensor on centrifugal pumps. It is measured with the help of vibration sensors (ADXL345), current sensors (SCT-013) and seal-leakage sensors. An ESP32 is used to monitor them in real-time. Then, particular information is uploaded to a cloud service where it is analyzed, and on the basis of a threshold, early warnings are produced. The system has vibration monitoring and current monitoring that should point to anomalies like bearing wear, overload and seal failures. The findings imply satisfactory fault detection, high accuracy, reduced downtime, and enhanced maintenance. Downtime and costs can be reduced by up to 30 per cent and 25 per cent, respectively, according to the industry case studies. The research gaps, though, can be explained by the fact that it is not tested under various operating conditions, and it is not properly integrated with other advanced machine learning models, nor is there an understanding of other parameters, including temperature and acoustic signals.

III. PROPOSED METHODOLOGY

The research methodology implies that the project will entail the design and development of an intelligent, real-time system of a pump failure detector that is based on an ESP32 microcontroller and multiple sensors and operates on an IoT platform. To achieve the desired operational parameters such as temperature, and flow rate that are important to the well-being of the pump, it is planned to get the required parameters at regular intervals. The ESP32 then compares sensor values to real-time values to determine abnormal trends, which can be an indication of common fault conditions, including overheating, cavitation and flow faults. The system conditions are categorized as normal or

abnormal, and this is done through a decision mechanism that works according to a threshold. The system also makes the pump ready to automatically shut down in case of abnormal behaviour, which minimizes the damage. In addition, one can track and observe data remotely when the integration of IoT is applied, which allows for preventive maintenance and increases the reliability of the system.

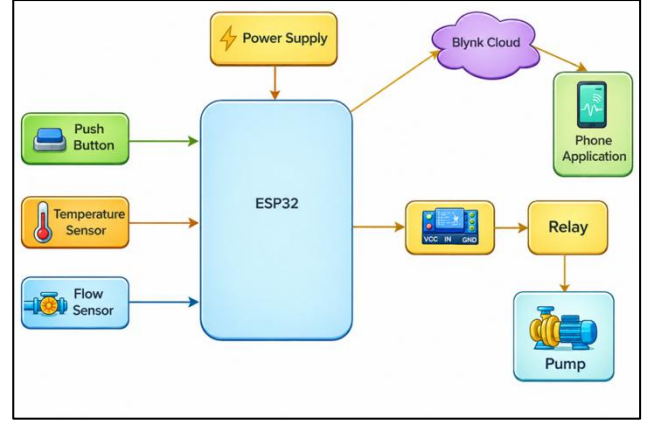


Fig.1. Block diagram of the proposed system.

The block graph gives the general outline of the proposed ESP32-based pump failure detection and monitoring system, and it indicates how the sensing, processing, communication, and control units are related to each other. The ESP32 microcontroller is the central node in the system, and it is the main processor. It takes the input signals of the various sources, which are the temperature sensor, and flow sensor, and constantly observes the appropriate operational parameters of the pump. The sensors would be able to provide real-time data regarding the health and performance of the system. It also possesses a manual control push button or an emergency button. This power supply unit ensures that all the components have a constant power supply, and this makes the system a steady and continuous process. The sensor data is then transmitted to the ESP32, which compares the sensor data with the predefined threshold values to conclude whether the pump is in a normal or abnormal condition. This remote monitoring setup features ESP32 and Blynk Cloud, which enables wired network transmissions of data. The processed data is transmitted to a mobile application where the client can be shown system parameters in real time and get notifications of abnormal conditions. Control side: The ESP32 on the control side will be connected with a relay module, which is an electromechanical switch. The reasoning behind the decision is that the relay will manage the current that passes to the pump. The pump is stopped by switching on the relay on ESP32 when any of the parameters is overrun so that the ESP32 remains intact and is not harmed. It is a closed system with a block scheme that comprises the sensing, processing, and IoT-based monitoring in addition to automated control that allows the successful and reliable detection of pump failure with minimal human intervention.

IV. EXPERIMENTAL SETUP AND CONTROL MECHANISM

The proposed pump failure detection system has a major attribute. One major attribute of the proposed pump failure detection system, as it will be controlled and set under the experimental conditions, is that the proposed system provides a high level of accuracy in data collection, stable operation of the system, and the absence of failures. It is

expected to simulate the environment of the centrifugal pump in reality with the assistance of needed hardware, including the pump, sensors, ESP32 microcontroller, and a relay module in a steady and well-designed format. In order to check the health of the pumps, several sensors in the critical locations will be provided to continuously check on the major parameters, including temperature, and flow rate. The physical installation will be provided alongside a smart control system that will have a predefined level, which will automatically react when an abnormal situation takes place. This is an experimental design and automated control system, which offers real-time observation, as well as the safety of the system, and the prevention of potential pump breakage beforehand.

A. Control Mechanism

The controlled conditions will be simulated conditions that will be equal to the real operating conditions, simulated conditions being the operation of the pump in real time. A recirculation system will be installed on the pump to make sure that the pump is not out of water; this will be in the form of pipes and a tank where water will be stored. The sensors will be properly located to get the correct information: the temperature sensor will be next to the motor of the pump will be on the pipeline, and the flow sensor will be in the flow of water. The system is fixed on a solid steel structure, which offers the mechanical stability of the whole system and proper positioning of all the parts during operation.

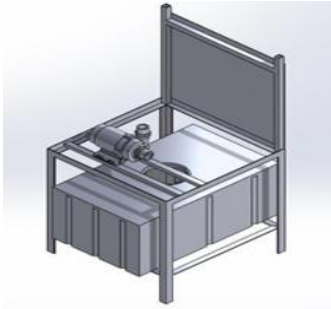


Fig.2. Control Mechanism

The 3D model of the given system will consist of the general architecture and design of the components of the pump monitoring system. It provides a nice image of the pump, sensors, piping system and frame. The model assists in getting to know the geographic structure and how the elements are united in achieving congruency and successful functioning. It can also be used to calculate the possible design limitations and enhance the system plan in general before actual implementation.

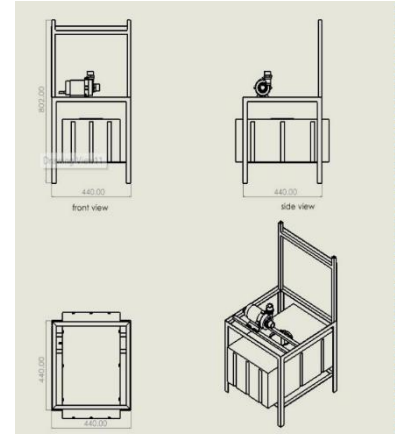


Fig 3. 2D design of system layout.

The 2D map of the system is the special layout and dimensional design of the elements needed in the setup of the experiment. It gives proper measurements and construction specifications of assembly machines. This design would enable the pump, sensor, and other elements that provide support to be fixed in the appropriate locations, which would make their installation and maintenance easier.

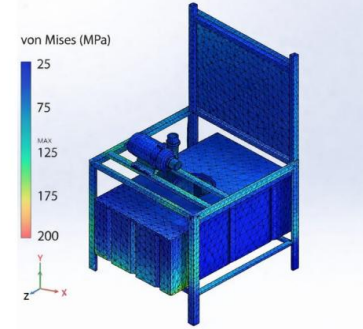


Fig 4. Finite Element Analysis of the system structure.

Analysis of the system structure to determine the level of mechanical strength and stability of the supporting frame during working conditions is done using a finite element analysis (FEA). This analysis demonstrates the redistribution and deformation of stresses in the structure, which is necessary to support the loads caused by the pump operation. The outcomes show that the chosen material and design are strong and non-hazardous, and allow using the system indefinitely without structural disintegration.

B. Control Mechanism

The monitoring mechanism of the system is based on the established set of threshold values of all parameters being monitored. The ESP32 constantly checks sensor data and decides whether the working conditions are safe. The pump will not be shut down even when the values taken are within a reasonable range. An ESP32, in its turn, requests the relay when any of the parameters reach its threshold, i.e. the device is in a fault state. This relay will then cut off the pump, which will then shut down to avoid further damage. Such a control system allows making the system safer and more reliable, because it is automated.

The benefits of the suggested system are as follows.

- Proposed system: The suggested system may be utilized to design a cost-efficient system for real-time monitoring and fault detection in pumps.

- A sensor-driven data gathering framework and ESP32 microcontroller will make the system record significant parameters and foul play in real-time.
- Characteristics: The benefits of the automatic shutdown feature are that they contribute to the enhancement of the safety of operation, and simultaneously, its small-scale design enables it to be used in the work of the small- and medium-sized plants.
- The system itself will also be able to monitor the performance of the pumps through the IoT, therefore allowing the user to plan the maintenance process more effectively, even when not being on-site.

V. RESULTS AND DISCUSSION

In the Results and Discussion section, the author describes how the proposed ESP32-based pump failure detection system worked under different operating conditions. It was established that the key parameters that characterize the behaviour of the pump were researched by means of the experimental measurements of voltage, current, power, power factor, and temperature. The acquired data was analyzed adequately and employed to measure the stability of the systems, track the accuracy and identify abnormal conditions. Besides that, the results demonstrate how the system responds to the change in operating parameters and the correctness of the threshold-based control mechanism. It is also in this section that the interrelation between the various parameters is made clear, and the ability of the system to identify when something is wrong at the initial stage enhances the efficiency and safety of the system.

A. Result analysis

In order to determine the functionality of the proposed pump monitoring system during the normal operating conditions, key parameters were measured, and they included current, power, voltage, power factor, and temperature. Observations were followed in a sequence, making an attempt to examine the behaviour of the pump and develop a baseline of how the pump should be operated. The pump is running with stability, as can be seen from the data received: the voltage is 220-240 V and 3.9-5.6 A. It has a power consumption of between 724 W and 1140 W, and a power factor of between 0.75 and 0.94.

The temperature was found to be between 38 °C and 53 °C, and this indicates that in normal conditions, the pump is operating within the normal range as far as thermo-performance is concerned. The readings can be used to act as a guide to abnormal conditions; any major deviation on these scales may provide evidence of faults or overheating and overloading. The fact that the data recorded was consistent implies that the implemented system to monitor operational parameters captures real-time operational parameters.

TABLE I. COMPARATIVE ANALYSIS

Sr. No.	Voltage (V)	Current (A)	Power (W)	Power Factor	Temperature (°C)
1	232.48	4.46	901.19	0.87	49.57
2	228.83	4.43	909.63	0.90	47.30
3	227.65	4.66	887.51	0.84	43.60
4	231.21	3.93	724.60	0.80	43.31
5	224.94	4.59	850.25	0.82	40.76
6	237.33	4.43	896.25	0.85	40.73

7	227.28	4.53	840.19	0.82	46.13
8	227.00	4.41	833.30	0.83	50.56
9	229.93	4.18	841.21	0.87	41.34
10	231.04	3.91	732.27	0.81	45.59

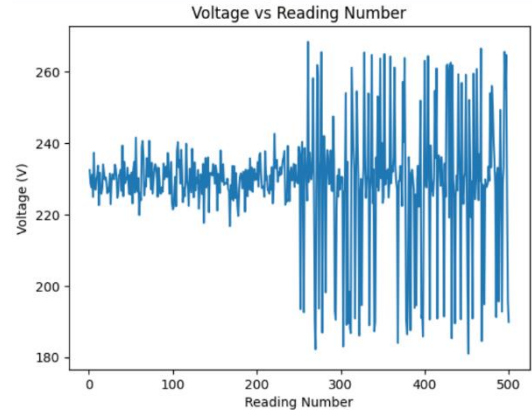


Fig 5. Variation of voltage under normal operating conditions

The voltage graph presents the regularity of the electric power when the pump is in operation. The voltage is, however, within a restricted range, and this implies that there is an unstable power input into the system. The loading variations are not very significant to the pump performance. The voltage value being constant depicts that the system is operating.

The graph below illustrates the variation in the current consumption of the pump. The values observed fall within the given range, which means that the load is normal. Small changes in current are linked to changes in the load and flow conditions of the pump. In case the current flows are high, there might be overloading or mechanical failure; when the current is kept at a steady rate, the pumps will work with efficiency and will work normally.

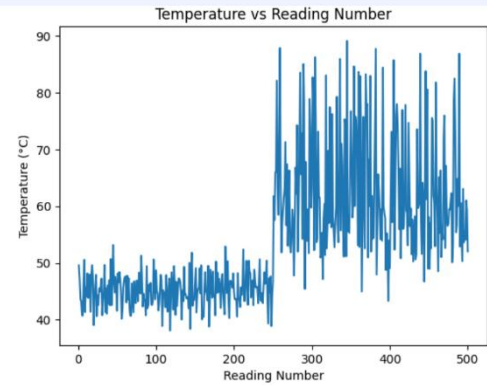


Fig 6. Temperature variation of the pump during operation

The temperature change graph will indicate the thermal behaviour of the pump. It is not too hot, hence it does not overheat and works well. This is found to be constant in the temperature variations due to constant activities and environmental influences. When the temperature is higher than the normal temperature, this would show that there is overheating or faults, and so temperature is a convenient parameter that can be used in fault detection.

B. Analysis of System Performance

The data were collected and analyzed to find out whether the proposed system can be effective in the identification of abnormal conditions. This was found to show that the system will be able to detect the variation in paramount parameters and act upon it. Indicatively, low temperature as compared to the normal range suggests overheating. On the other hand, pressure deviation and flow rate can be signs of cavitation or clogging. It is accomplished by recording sensor data against the set limits by the system until the deviations are noted. Laws at various parameters also help to understand the behaviour of the pump. Normal progression of increased current and power consumption is linked with a high load state that can eventually result in overheating. In the same manner, the reduction in flow rate, or any variation in pressure, could be a sign of a blockage or a leak in the system. These relations of the parameters contribute to the system to perform better fault detection and give early warnings to avoid critical failures.

C. System Response to Fault Conditions

The suggested system has been open to abnormal conditions. In the case where one of the parameters surpasses the set parameter, the ESP32 microcontroller compares the data to the set parameter and the relay module is switched off automatically to cut off the pump. An automatic reaction system like this will aid in averting any more harm to the pump and keeping it in safe use. The ability to track the work of the pumps and monitor their performance, as well as check them in real time, allows the tracking of their work at any time, and their connection to the IoT makes it possible to monitor them remotely. It will make the conditions of the system monitorable and provide a possibility to take the required actions even in the absence. The speed of the reaction and reliability of the system prove the value of its applicability in the practical setting, both in the industrial and domestic settings.

D. Discussion

The conducted experiment results suggest that the suggested monitoring system, which is based on an ESP32, can be effectively employed to inform about pump failures promptly. The system is also effective in monitoring important parameters and detecting deviant trends that reflect the possibility of failure. Compared to the traditional maintenance systems, this system is proactive, monitored and controlled in real-time. The sensor-based monitoring system will be employed to minimize the use of manual searches and minimize the presence of unpredictable failures. The system is also affordable as a low-priced microcontroller, such as the ESP32, which can be deployed to small and medium-scale applications. Remote monitoring and data analysis are other applications of the Internet of Things. The system has a high level of operationality, less maintenance is needed, and it is very reliable. The findings indicate the efficiency of the presented predictive maintenance and safe pump functioning.

VI. CONCLUSION AND FUTURE SCOPE

This article presents a study of a pump failure detection & monitoring device that was designed and constructed using an ESP32 microcontroller. An attached temperature and flow measurement device was also designed to allow for the pump to operate smoothly and reliably by providing constant real-time feedback from 3 different sensors. The pump detection

system used these 3 measurements, successfully detecting when an impending failure may occur. The testing data demonstrates that the pump failure detection system was able to provide information to define what each measurement means to the operational metrics of the pump. A proposed method for troubleshooting this problem is to install control mechanisms & set thresholds within the pump system that will automatically shut down the pump when it is out of limits (like 10° C) to protect both the pump from increased wear and improve safety in the overall system. In summary, the system can be put in place under standard operating conditions and is capable of tracking any deviations from normal operating ranges, therefore helping to minimize the probability of failure.

The ESP32 offers a low-cost and accessible approach to monitoring in real-time, along with integrated Wi-Fi capabilities, which will allow for Internet of Things (IoT) based remote monitoring. This system is much more proactive than traditional maintenance methods; because it provides the ability to detect faults in advance, it can reduce unplanned outages by minimizing or eliminating downtime. In addition to improving the reliability of the pumps themselves, this system will reduce the costs associated with ongoing maintenance and improve the overall performance of the pumps. Despite the fact that the system has been successful to date, there are still opportunities to increase its functionality and usability. Future work will provide opportunities to improve predictive capabilities through the use of more advanced machine learning and artificial intelligence techniques, which will lead to improved prediction quality and decision-making capabilities with respect to risks in pump systems. Future work could also add additional sensors to this system, such as vibration measurements, to provide for more in-depth analyses of the faults experienced by pumps.

The future phase of advancement may involve the utilization of cloud storage and IoT-enhanced dashboards to visualize and explore data instantaneously, as well as monitor pump efficiency. This would enable increased levels of quality and intelligent maintenance for multiple pumps. Furthermore, this is optimally designed for industrial applications and can work in conjunction with the Fourth Industrial Revolution to augment the system's features. The current approach will also apply to other industrial machinery, making this an adaptable intelligent maintenance system framework. Additionally, these advanced technologies will create future monitoring systems with more advanced capabilities.

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